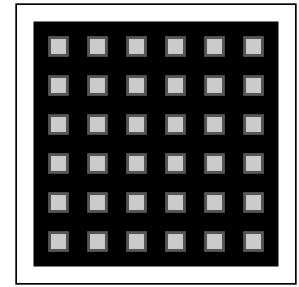


On the Subject of RGB Tree

How could the root node be at the top layer and the leaf nodes be at the bottom layer? It's supposed to be a tree! This makes no sense...



The 6x6 grid of buttons forms a quaternary-tree containing 36 pressable nodes.

Each node (except one) points to a neighboring node directly above, below, left, or right of it. These arrows always point one step closer to the "root" node.

The root is the single node with no pointer, and every path in the tree eventually leads to it.

The tree contains no loops, and nodes are never isolated.

Nodes that are never pointed to by another are "leaf" nodes.

Each leaf holds an integer (1, 2, or 4) in binary, specifically in three bits.

Bits represent color channels. Bit 1 = Red, Bit 2 = Green, and Bit 3 = Blue.

If a bit is equal to 1, its color channel is activated for the button's light.

A leaf will send its integer to every node in its path towards the root, applying an XOR operation to the node's starting integer 0.

If a yellow-lit (011) and a red-lit node (001) led to the same button, the button's XOR'd value will be 010, making it green-lit.

Leaves are the red, green, and blue buttons that are more dimly lit than the rest. Clicking a leaf node will shuffle the integers of all leaves (between 1, 2, and 4), which is helpful for mapping out the tree structure.

To engage submission, find and click the root node.

Find any node that is the farthest from the root (in steps along the tree's paths).

Then starting from the root, click adjacent nodes step-by-step, tracing a valid path until you reach and click the far node, solving the module.

If multiple nodes share the same max. distance, any of them are valid targets.

You are allowed to move forwards or backwards through the tree's paths.

Initially failing to click the root node will incur a strike.

Failing to follow a valid path through the tree will also incur a strike, and cancel submission.

If the tree cannot be fully deduced just by using the lights, see next page.

The tree may be impossible to deduce due to snake-like paths.
If and ONLY if this is the case, a new mode will be enabled.

Each node that cannot be deduced, as well as all leaf nodes, will be lit up based on which direction their arrows point to.

A random color is chosen for each direction (above, below, left, and right).

If the root node is among the list of ambiguous nodes, it will be given a random one of these four colors.

To enter this mode, click every leaf node in a specific order; scan rows left-to-right, moving through rows from bottom-to-top.

You cannot submit while in this mode. To exit, click any leaf node again.

Trying to engage in this mode when the tree can in fact be fully deduced will incur a strike.

*If using colorblind mode, white-lit and off buttons are not labelled, and leaf nodes are labelled with white text.